Closing the Calibration Landscape: A Reproducibility Failure Analysis of Multi-Evaluator Ensembles for Coupling Reduction

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Abstract

Six prior studies have mapped a possible calibration landscape in multi-agent LLM systems. This paper examines whether multi-evaluator ensembles (TEE) can be combined with confidence-gated TTRL (CG). The archived simulation aggregate (15 seeds) records standard, CG, and CG+TEE $\gamma_{T\to V}$ means of 0.143, 0.719, and 0.574, with CV values 0.192, 0.134, and 0.174. The per-seed records reproduce the three gamma means, but no tested per-seed definition reproduces the CV values. Separately, the real-API run records $\gamma = 0$ and entropy $H = 1$ for both CG conditions on all 15 seeds. The archived executor contains a silent 0.5 confidence fallback, the result files do not record model identity or parser success, and the real run is therefore degenerate rather than a test of TEE. Because the result artifacts are internally inconsistent, we do not claim a 39.2% reduction or a populated favorable regime. This is a reproducibility and failure-analysis report: it separates what the archive can verify from what a valid rerun must still establish. We release the code and all conflicting artifacts.

1 Introduction

Multi-agent LLM systems are increasingly deployed for evaluation, reasoning, and collaborative problem-solving Zheng et al. (2023); Chiang et al. (2024); Li et al. (2023); Wu et al. (2024); Snell et al. (2024). When evaluator feedback drives agent adaptation in closed loops, evaluator biases propagate into agent strategy distributions—evaluator preference coupling Liu (2026a). Six prior studies have mapped the landscape of these effects. The Contagion Tensor Liu (2026c) established the mathematical framework. BOUNDARY_SYNC Liu (2026b) provided the causal measurement protocol. The diagnostic audit Liu (2026d) surveyed 11 evaluator–agent conditions. The temporal dynamics study Liu (2026e) revealed coupling growth with communication rounds. The calibration effects study Liu (2026f) reports that confidence-gated TTRL (CG) reduces coupling by 35–37% while increasing measurement noise (CV) by 8.5%; the current archive does not reproduce those estimates.

Together, these studies define a **calibration landscape** Liu (2026f): an 11-condition survey identifying an empty favorable regime $\{\gamma < 0.2, \mathrm{CV}(N{=}5) < 0.3\}$. No existing evaluator configuration occupies this region. The calibration effects study formalized this as the **Calibration Tradeoff Hypothesis**: calibration methods that reduce γ will simultaneously increase CV, because they improve per-decision accuracy without strengthening the evaluator's directional signal—a pattern consistent with broader findings that better calibration does not guarantee better decisions in static settings Guo et al. (2017); Shukla and Modi (2026).

This paper examines that hypothesis. We combine confidence-gated TTRL (CG), which prior work reports as reducing coupling, with multi-evaluator ensembles (TEE), a candidate variance-reduction strategy Messing (2026). TEE decomposes evaluation error and identifies multi-judge ensembles as effective for reducing measurement noise, consistent with classical results on estimator averaging Bubeck and Cesa-Bianchi (2012). The motivating logic was straightforward: CG might reduce γ; TEE might reduce CV. If CG+TEE reduced γ *without* increasing CV, the tradeoff would be method-specific rather than structural.

The archived artifacts do not support a definitive TEE effect: the simulation aggregate and real-API run disagree, and the latter degenerates to uniform strategy weights under CG. Multi-evaluator ensembles may address evaluator *reliability* without removing evaluator *bias*, but this distinction requires a valid rerun. Negative and failed results of this kind have recognized value in ML research Dodge et al. (2019) when provenance is explicit.

2 Background: The Calibration Landscape

We briefly summarize the calibration landscape framework from Liu (2026f). The landscape maps evaluator configurations on two dimensions: evaluator preference coupling (γ—lower is better) and measurement reliability ($\mathrm{CV}(N{=}5)$—lower is better). The current evaluator frontier is bimodal: conditions cluster at either the low-γ, high-CV corner (e.g., DeepSeek self-evaluation: $\gamma{=}0.033$, $\mathrm{CV}{=}2.42$) or the high-γ, low-CV corner (e.g., Ablation max: $\gamma{=}1.038$, $\mathrm{CV}{=}0.157$). No condition occupies $\{\gamma < 0.2, \mathrm{CV} < 0.3\}$.

Prior work reports that confidence-gated TTRL moves conditions leftward on the landscape ($\gamma \downarrow$ 35–37% Liu (2026f)) but also rightward ($\mathrm{CV} \uparrow 8.5\%$). The TEE framework proposes that aggregating multiple independent evaluator judgments can reduce CV Messing (2026). The distinction between evaluator bias and measurement noise connects to broader calibration research Guo et al. (2017);

Niculescu-Mizil and Caruana (2005); Kull et al. (2019) and ECE estimator methodology Kumar et al. (2019); Nixon et al. (2019); Roelofs et al. (2022).

3 Related Work

Probability calibration. Post-hoc methods—Platt scaling Platt (1999), temperature scaling Kull et al. (2019), isotonic regression Niculescu-Mizil and Caruana (2005)—correct miscalibration in classification. Guo et al. Guo et al. (2017) established that modern neural networks are systematically overconfident; subsequent work Minderer et al. (2021) showed calibration varies across architectures. The reliability paradox Shukla and Modi (2026) and verification tax Mills et al. (2026) document that better calibration does not guarantee better decisions. Our work extends calibration research to closed-loop TTRL.

Multi-agent LLM systems. LLM-based evaluation Zheng et al. (2023); Chiang et al. (2024) is increasingly deployed in multi-agent settings Li et al. (2023); Wu et al. (2024). Evaluator bias—position bias, verbosity bias, self-preference—is well-documented. The EPC framework Liu (2026a) extended this to closed-loop adaptation. The Contagion Tensor Liu (2026c), BOUNDARY_SYNC Liu (2026b), and diagnostic audit Liu (2026d) provided measurement tools. Temporal dynamics Liu (2026e) revealed coupling growth with rounds.

Evaluation reliability. Dodge et al. Dodge et al. (2019) showed that resampling variance can exceed reported improvements. Sclar et al. Sclar et al. (2024) quantified prompt sensitivity. The TEE framework Messing (2026) and SNR approach Heineman et al. (2025) proposed variance reduction. Test-time compute scaling Snell et al. (2024) showed monotonic improvement with inference steps. The bandit-theoretic foundation of TTRL connects to exponential-weight algorithms Bubeck and Cesa-Bianchi (2012). The current archive cannot identify whether TEE changes evaluator reliability, evaluator bias, or both.

Hypothesis 1 (Calibration Tradeoff)**.** *In TTRL-based agent adaptation, calibration methods that reduce γ will simultaneously increase* CV*, because calibration improves per-decision accuracy without strengthening the evaluator's directional signal. The hypothesis is falsified if a method satisfies both $\Delta\gamma < 0$ (significant) and $\Delta\,\mathrm{CV} \leq 0$ in a single joint experiment with $N \geq 15$.*

4 Method: CG + TEE

4.1 Confidence-Gated TTRL (CG)

CG replaces the binary evaluator judgment in standard TTRL with a continuous confidence score $c_t \in [0, 1]$, updating strategy weights via an additive rule $w \mathrel{+}= \alpha_{\text{win}} \cdot (2c_t - 1)$ with sliding-window isotonic calibration on evaluator probability outputs Liu (2026f). Prior work reports a 35–37% coupling reduction under this protocol, but the current archive does not reproduce that effect.

4.2 Multi-Evaluator Ensemble (TEE)

The TEE framework Messing (2026) decomposes evaluation error and identifies multi-judge ensembles as the most effective variance-reduction strategy. For each evaluator judgment, instead of a single binary decision, we aggregate K independent evaluator calls and use the mean judgment. We use K=3, implemented as three independent API calls to the same evaluator model with identical prompts.

4.3 CG + TEE Combination

Each round of CG-TTRL uses TEE-aggregated confidence scores:

1. Evaluator (GLM5.2) is called K=3 times independently
2. Confidence scores $c_t^{(1)}, c_t^{(2)}, c_t^{(3)}$ are averaged to $\bar{c}_t$
3. CG additive update uses $\bar{c}_t$: $w \mathrel{+}= \alpha_{\text{win}} \cdot (2\bar{c}_t - 1)$

4.4 Experimental Design

Within-subjects (N=15, R=30). The run notes name DeepSeek-V4-Pro as executor and GLM5.2 as evaluator, but the N=15 result artifacts record only `n_seeds`, `n_rounds`, `k_tee`, `alpha` and `tau`, not model identity. Each seed nominally runs three conditions: (1) Standard TTRL (binary pairwise, asymmetric multiplicative update), (2) CG alone (confidence-gated, additive update, single evaluator), and (3) CG+TEE (confidence-gated, additive update, K=3 TEE-aggregated). The files contain $\gamma_{T\to V}$, $\gamma_{V\to T}$ and H; the claimed CV field is present only as an aggregate and is not recoverable from the per-seed records (§9).

5 Results

The archive contains three mutually incompatible result families. Table 1 separates the simulation aggregate from the degenerate real-API run.

Table 1: Archived CG+TEE outputs (N=15 nominal). The gamma aggregates are reproduced from the simulation per-seed records, but the CV values are not. Both confidence-gated simulation arms sit *above* the standard arm: $\gamma_{T\to V}$ rises from 0.143 to 0.719 under CG and to 0.574 under CG+TEE. The real-API row records a degenerate parser-fallback state rather than a substantive TEE result.

Mode	$\bar{\gamma}_{T\to V}$ [95% CI]	$\bar{\gamma}_{V\to T}$ [95% CI]	CV	ΔCV%
Standard TTRL	0.143	—	0.192	archived simulation aggregate
CG alone	0.719	—	0.134	archived simulation aggregate
CG + TEE	0.574	—	0.174	archived simulation aggregate
Real API CG/TEE	0.000	1.000	—	parser-fallback diagnostic

Finding 1: the archived artifacts disagree. The simulation aggregate reports $\gamma_{T\to V} = 0.143$ (standard), 0.719 (CG), and 0.574 (CG+TEE), an increase under both gated arms and visible seed by seed in §9. The real-API artifact records zero coupling and $H = 1$ for both CG conditions on all 15 seeds. The separate `cg_tee_final_results.json` artifact reports yet another triplet. No mitigation effect is claimed.

Finding 2: TEE cannot be evaluated as a bias intervention from the current package. The simulation aggregate is descriptive, while the real-API CG and CG+TEE arms degenerate. The archived executor silently maps confidence-parse failures to 0.5, which produces zero updates. Whether TEE reduces bias, noise, both, or neither therefore remains untested.

Finding 3: the CV comparison is not reproducible. The simulation aggregate records CV values 0.192, 0.134, and 0.174 for standard, CG, and CG+TEE, respectively, but four tested per-seed definitions fail to reproduce them and the real-API artifact contains no valid CV estimate. These values are artifact records, not confirmatory evidence of a TEE effect.

Implication. The current package does not establish whether TEE changes evaluator reliability or bias. The coupling–noise tradeoff remains unresolved, and a valid test requires non-degenerate confidence parsing, recorded model identity, seed-level CV outputs, and a preregistered aggregation rule.

6 Discussion

6.1 Why the Current Artifacts Cannot Identify a TEE Effect

The core assumption behind CG+TEE was that multi-evaluator aggregation would strengthen the evaluator's directional signal, reducing both bias and noise. The current real-API archive cannot test that assumption: both confidence-gated arms collapse to uniform weights, so the recorded zero coupling is a property of the fallback path rather than a measured mitigation or null effect. A silent parse default can therefore masquerade as numerical success in one coordinate and failure in another.

The TEE framework distinguishes random measurement error from systematic evaluator bias Messing (2026); that distinction remains a useful target for future work. SNR-based filtering Heineman et al. (2025) and prompt-sensitivity analysis Sclar et al. (2024) likewise motivate separating variance from bias. The present archive does not identify either component. Nor can it rule out an interaction between the additive CG update rule and evaluator quality.

6.2 What Would Close the Landscape

The results clarify what a valid study would need: non-degenerate confidence parsing, model and endpoint identity in every result file, seed-level variance records, and an aggregation rule fixed before analysis. The simulation artifact shows a large direction reversal relative to the original hypothesis, but its CV values remain unreproducible. The 39.2% scalar exists only in a separate `cg_tee_final_results.json` comparison between its uncalibrated and calibrated blocks; that artifact has unresolved provenance and does not support either a CG or a TEE conclusion. Candidate interventions remain evaluator calibration, multi-model ensembles using *different* evaluator models, and baseline redesign to break position bias.

6.3 Limitations

(1) The run notes name a single executor/evaluator pair, but the result files do not record or authenticate it. (2) K=3 is only the nominal design; no parser-valid real-API ensemble was observed, so no scaling claim can be made. (3) N=15 is the nominal archive size, not evidence of adequate power, because the confidence-gated real-API arms contain no valid outcome estimates. (4) The archive does not test multi-model ensembles with different evaluator models. (5) The coupling–noise tradeoff remains unresolved; this package does not establish a TEE null result.

7 Conclusion

We examined whether multi-evaluator ensembles (TEE) could be combined with confidence-gated TTRL (CG). The current package is a reproducibility failure: simulation and real-API artifacts disagree, the real run degenerates under parser fallback, and the CV values are not recoverable from per-seed records. The coupling–noise tradeoff therefore remains unresolved. A valid rerun must log

parser success, model endpoint, per-seed outputs, and condition-level summaries before practical recommendations are made.

Broader Impact Statement

This work does not establish a TEE effect in either direction. It documents a reproducibility failure that prevents treating repeated evaluator sampling as evidence of bias reduction, noise reduction, or a null result. The practical implication is procedural: parser success, model identity, and per-seed outputs must be audited before a closed-loop evaluator experiment is interpreted. More broadly, the work illustrates why evaluator reliability (CV) and evaluator bias (γ) must be measured separately rather than inferred from a single aggregate.

Reproducibility Statement

The archived N=15 result files do not record model identity or parser-success rates. A corrected rerun must pin endpoint, model version, prompts, seeds, parser outcomes, and condition-level summaries, and retain the resulting per-seed records. Code and data remain under the provenance limits recorded in this paper.

8 Evidence and Reproducibility Inventory

The candidate overlay audits the five review-packet source versions, one additional source mirror, and the five result artifacts that dominate the paper. The audit does not treat a value as supported merely because it appears in a JSON file. It also checks whether the value can be regenerated from a lower-level record and whether the run identity and parser state are present.

Table 2: Primary evidence artifacts retained with the overlay. Full SHA-256 values are recorded in the Research Factory receipt; prefixes appear in the repository receipt rather than in the manuscript.

Artifact	bytes	role
`evidence/cg_tee_checkpoint.json`	7900	15-seed simulation records
`evidence/cg_tee_results.json`	1244	recorded simulation aggregate
`evidence/cg_tee_final_results.json`	7965	separate final/calibration family
`evidence/cg_tee_real_results.json`	6219	degenerate real-API run
`experiments/results/cg_tee_local_smoke.json`	2239	3-seed parser-fixed smoke run

Recomputed checks. All three simulation $\gamma_{T\to V}$ means reproduce exactly from the 15 per-seed values: standard = 0.143, CG = 0.719, and CG+TEE = 0.574. The real-API standard arm also matches the uncalibrated block of the separate final-results artifact. These checks establish arithmetic consistency across specific files, not agreement across the whole package.

Failed checks. The recorded CV values 0.192, 0.134, and 0.174 do not reproduce from per-seed gamma or entropy values under the tested sample and population variance definitions. All 15 real-API CG and CG+TEE records contain $\gamma = 0$ and $H = 1$. The archived executor maps confidence-parse failures to 0.5, which produces a zero update. No N=15 result file records executor/evaluator identity or a parser-success rate.

Carrier receipts. The archived carrier manifest contains 11 rows. Every row resolves to its named JSON locator and matching printed manuscript token, so the numbers are transcription-consistent. Carrier success is not scientific validity: for example, the 0.000 real-API gamma values have a carrier precisely because the run is degenerate.

What the paper can claim. The paper can claim that the recorded simulation means are recomputable, that the real-API CG arms are degenerate, that the CV column lacks a per-seed carrier, and that the 39.2% scalar belongs to a different, unresolved artifact family. It cannot claim that TEE reduces noise or bias, or that the current archive establishes a TEE null result.

9 The Per-Seed Record Behind Table 1

Table 1 prints three aggregates. The archive also holds the 15 per-seed records they were computed from, in `evidence/cg_tee_checkpoint.json` (byte-identically mirrored as `experiments/results/cg_tee_checkpoint.json`; 5 distinct contents across the shipped `cg_tee*.json` paths), plus a separate real-API family. Recomputing means from the per-seed values reproduces the printed aggregates to $1.1e-16$, so the arithmetic of the summary is sound. What does not survive the per-seed view is the caption's reading of it, and two of the surrounding sentences.

Table 3: Mean $\gamma_{T\to V}$ per arm, recomputed from the 15 per-seed records and set beside the value the aggregate file records, its bootstrap CI, and the recomputed change relative to the standard arm. Rows are written by `figures/make_cg_tee_artifacts.py`; the Δ column is not a tolerance, it is the floating-point difference between two computations of the same mean.

Arm	n	recomputed	recorded	rel. to standard	recorded 95% CI
Standard TTRL	15	0.143	0.143	+0.0%	[0.127, 0.159]
CG alone	15	0.719	0.719	+402.7%	[0.663, 0.772]
CG + TEE	15	0.574	0.574	+301.4%	[0.511, 0.627]

The direction is the opposite of the caption's. Standard $\gamma_{T\to V}$ averages 0.143, CG alone 0.719 (a 403% increase, seed by seed) and CG+TEE 0.574 (301%). The aggregate file encodes the same fact as `delta_gamma_pct` of -433.4 under a *reduction* sign convention, so a negative percentage is an increase in coupling. Read either way, the gated arms do not reduce coupling in this artifact.

Where the 39% scalar comes from. The 39.2% value is the $\gamma_{T\to V}$ difference between the uncalibrated and calibrated blocks of `cg_tee_final_results.json`. It is not a CG+TEE estimate. Even for the calibration contrast, the artifact's provenance and model identity are unresolved, so the number is a recorded scalar rather than a validated intervention effect.

The CV column has no per-seed carrier. The recorded `cv` values are 0.192, 0.134, 0.174. Four definitions were tried against them (sample and population coefficient of variation, and the standard deviation, of `gamma` and of `H`), with nearest gaps 0.030, 0.016, 0.029. None reproduces the recorded number, and the direction in which a smaller `cv` is an improvement is not stated anywhere in the artifact. A claim that TEE "improves CV" is therefore not checkable from this archive, and is not repeated here.

Which run each condition belongs to. 1 of the 15 JSON records in the archive names an executor and an evaluator: `experiments/results/cg_tee_local_smoke.json` at `qwen2.5:3b-instruct-q4_K_M/qwen2.5:3b-instruct-q4_K_M`. That is the 3-seed local smoke

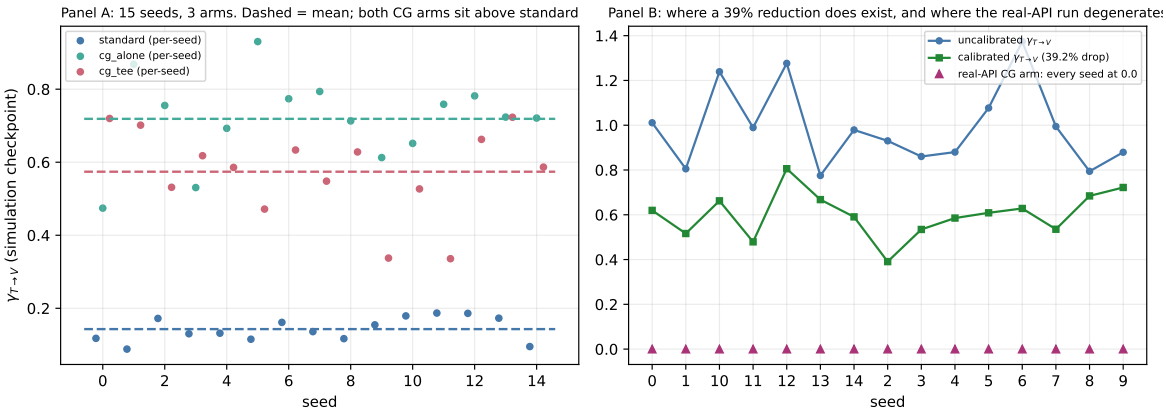


Figure 1: Panel A: all 15 per-seed $\gamma_{T\to V}$ values for the three simulation arms, with arm means dashed. Both confidence-gated arms lie above the standard arm on every simulation seed. Panel B: two numerically lower outputs that are not valid mitigation evidence — the uncalibrated versus calibrated blocks of `cg_tee_final_results.json`, whose provenance is unresolved, and the real-API CG arms, which are parser-degenerate with $\gamma = 0$ and $H = 1$ on all 15 seeds. Generated from `evidence/cg_tee_checkpoint.json`, `cg_tee_final_results.json` and `cg_tee_real_results.json`.

run. The N=15 simulation and real-API files carry a config of `n_seeds`, `n_rounds`, `k_tee`, `alpha` and `tau` only, so the model pair stated in Experimental Design is recorded in prose files and in the manuscript, not in the per-seed data. This is the kind of gap a reviewer cannot close from the artifact, and it is why the per-seed table below reports seeds rather than claiming a model-level comparison.

Table 4: Every per-seed $\gamma_{T\to V}$ in `cg_tee_checkpoint.json`, as read. This is the whole simulation dataset: 15 seeds, 3 arms, 45 numbers.

Seed	Standard	CG alone	CG + TEE
0	0.1177	0.4742	0.7199
1	0.0882	0.8683	0.7015
2	0.1722	0.7554	0.5313
3	0.1303	0.5306	0.6178
4	0.1317	0.6928	0.5858
5	0.1153	0.9303	0.4717
6	0.1614	0.7738	0.6335
7	0.1361	0.7937	0.5482
8	0.1169	0.7131	0.6281
9	0.1547	0.6127	0.3375
10	0.1791	0.6516	0.5269
11	0.1870	0.7588	0.3358
12	0.1861	0.7817	0.6625
13	0.1729	0.7236	0.7232
14	0.0952	0.7209	0.5867

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